

3 (**short**) Figure 3 shows the displacement diagram for a pin-jointed truss comprising five nodes, drawn to scale. Lines representing displacements due to bar extensions are solid, and those representing displacement due to bar rotations are dashed. Nodes A and B are restrained horizontally and vertically, and node B is horizontally a distance L to the right of A. Node C is vertically above node B. The only external force applied to the structure is at node E. The structure comprises horizontal and vertical members of length L and diagonal members of length $\sqrt{2}L$, all of which have the same Young's modulus E and cross-sectional area A .

(a) Draw the structure. [6]

(b) Find all the bar forces within the structure and the external force acting on it. [4]

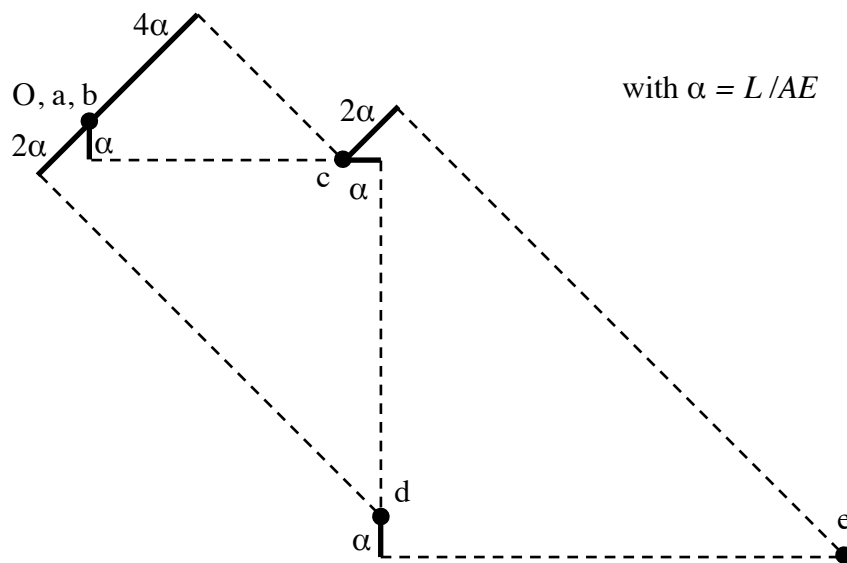


Fig. 3